

Requirement Analysis Document

Online Book Store System

Prepared by: Muhammad Shayan Haider | 10Pearls Shine Internship Program – QA Track (Cohort 9)

1. Project Overview:

This document presents a requirement analysis for an Online Book Store system, developed as part of the QA internship requirement analysis assignment. The purpose of this document is to translate the business needs of an e-commerce book platform into structured requirements, user stories, and acceptance criteria that can guide both development and testing efforts.

2. Requirement Understanding:

The Online Book Store is an e-commerce platform that allows users to browse, search, and purchase books online. The system aims to provide a seamless shopping experience by allowing users to explore a wide catalog, view detailed book information, manage a shopping cart, and complete secure purchases. Registered users can also manage their accounts and track their order history.

Key workflows expected of the platform:

- A visitor can browse or search the book catalog without creating an account.
- A visitor can view detailed information about a specific book.
- A visitor can add books to a shopping cart.
- A user must register or log in before completing a purchase.
- A logged-in user can check out, pay, and receive an order confirmation.
- A registered user can manage their profile and view past orders.

3. Functional Requirements:

Functional requirements describe what the system should do. The following requirements have been identified for the Online Book Store system:

- ✓ **FR1:** The system shall allow new users to register using a valid email address and password.
- ✓ **FR2:** The system shall allow registered users to log in using valid credentials and shall reject invalid login attempts with an appropriate error message.
- ✓ **FR3:** The system shall allow users to search for books by title, author, or category.
- ✓ **FR4:** The system shall display book details — including title, author, price, description, and availability — when a user selects a book.
- ✓ **FR5:** The system shall allow users to add or remove books from a shopping cart.

- ✓ **FR6:** The system shall allow users to complete checkout by entering shipping details and processing payment.
- ✓ **FR7:** The system shall send an order confirmation to the user's registered email address after a successful purchase.

4. Non-Functional Requirements:

Non-functional requirements describe the quality attributes of the system rather than specific features. The following have been identified:

Category	Requirement
Performance	The system shall load the homepage and search results within 3 seconds under normal load conditions.
Security	The system shall encrypt user passwords and payment information using industry-standard encryption (e.g., AES, TLS).
Usability	The system shall provide an intuitive interface that a first-time user can navigate without external instructions.
Availability	The system shall maintain at least 99.5% uptime, excluding scheduled maintenance windows.
Scalability	The system shall support at least 1,000 concurrent users without noticeable performance degradation.

5. User Stories:

User stories describe system interactions from the perspective of the user, using the format: "As a [role], I want [goal], so that [benefit]."

- ✓ As a **visitor**, I want **to search for books by title or author**, so that I can quickly find the book I am looking for.
- ✓ As a **visitor**, I want **to view detailed information about a book**, so that I can decide whether to purchase it.
- ✓ As a **registered user**, I want **to add books to my cart**, so that I can purchase multiple books in a single order.
- ✓ As a **new user**, I want **to create an account**, so that I can save my details and track my orders.
- ✓ As a **registered user**, I want **to log in securely**, so that I can access my account and order history.
- ✓ As a **registered user**, I want **to complete checkout with payment**, so that I can finalize my book purchase.
- ✓ As a **registered user**, I want **to receive an order confirmation**, so that I know my purchase was successful.

6. Acceptance Criteria (Gherkin Syntax):

The following scenarios define acceptance criteria for the core features of the system, written in Given-When-Then format:

Scenario: Search for a book by title

Given the user is on the bookstore homepage
When the user enters a valid book title in the search bar
Then the system should display a list of matching books

Scenario: View book details

Given the user has searched for a book
When the user clicks on a book from the search results
Then the system should display the book's title, author, price, and availability

Scenario: Add a book to cart

Given the user is viewing a book's details page
When the user clicks "Add to Cart"
Then the book should appear in the user's shopping cart

Scenario: Register a new account

Given the user is on the registration page
When the user enters a valid email and password
Then the system should create a new account and redirect the user to the login page

Scenario: Login with invalid credentials

Given the user is on the login page
When the user enters an incorrect password
Then the system should display an error message and not log the user in

Scenario: Successful checkout

Given the user has items in their cart and is logged in
When the user completes the checkout process with valid payment details
Then the system should confirm the order and send a confirmation email

7. Conclusion:

This document has translated the business goals of the Online Book Store system into functional requirements, non-functional requirements, user stories, and Gherkin-based acceptance criteria. These artifacts provide a clear and testable foundation for subsequent test case design and sprint planning activities.